

# Multimodal Satellite Imagery Regression Project

## 1. Project Overview

This project focuses on predicting a continuous target variable using a multimodal learning approach that combines satellite imagery data with tabular metadata (such as latitude, longitude, and other numerical features). The motivation is to leverage both visual spatial information and structured numerical data to achieve higher predictive accuracy than using either modality alone.

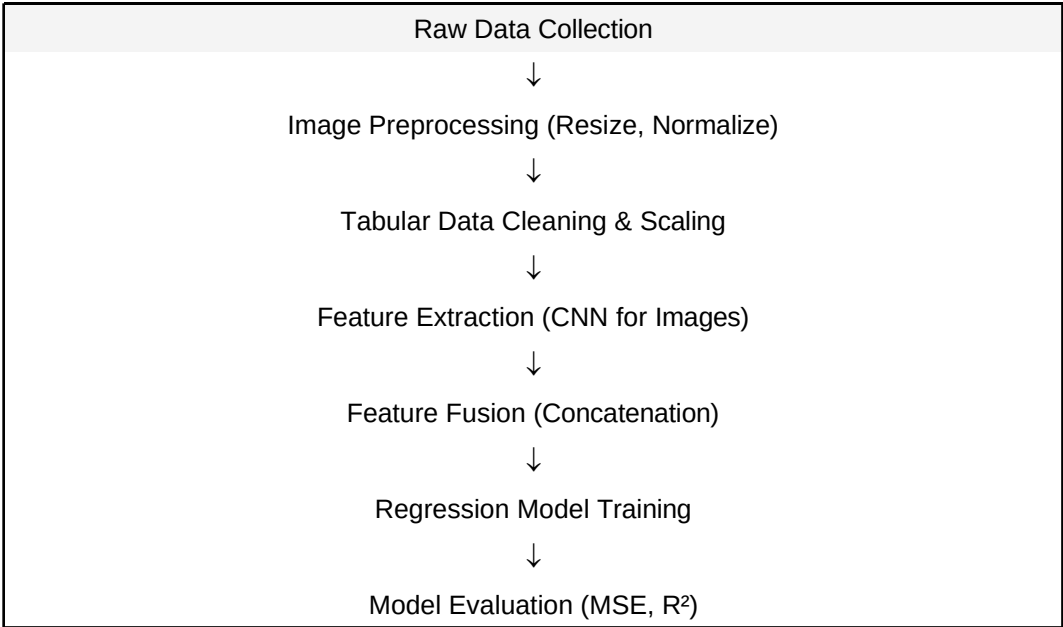
## 2. Problem Statement

Traditional machine learning models often rely only on tabular data, while deep learning models focus on image-based features. However, satellite-based prediction tasks benefit from both sources. The challenge addressed here is to design an end-to-end pipeline that efficiently fuses image features with tabular features and trains a regression model capable of generalizing well to unseen data.

## 3. Dataset Description

The dataset consists of satellite images stored as TIFF files and a corresponding CSV file containing tabular attributes and the regression target. Each image is uniquely identified by an ID that maps it to the appropriate row in the tabular dataset.

## 4. Process Flow



## 5. System Architecture

| Satellite Image                 | Tabular Features   |
|---------------------------------|--------------------|
| CNN Backbone                    | MLP / Dense Layers |
| Image Embedding                 | Tabular Embedding  |
| Feature Concatenation           |                    |
| Fully Connected Regression Head |                    |

|                             |  |
|-----------------------------|--|
| Predicted Continuous Output |  |
|-----------------------------|--|

## 6. Training and Evaluation

The model is trained using a supervised regression objective, typically Mean Squared Error (MSE). The dataset is split into training and validation sets. Performance is evaluated using metrics such as MSE and R-squared ( $R^2$ ) to measure prediction accuracy and variance explanation.

## 7. Conclusion

This multimodal architecture effectively integrates satellite imagery and structured data, resulting in improved predictive performance.

## 8.EXTRACTED IMAGES.

